

Assessment Scheme for Type 2-Projects

Project 1: Handwritten Digit Recognition

1. Model Implementation (20 marks)

(a) k-Nearest Neighbors (7 marks)

- Implementation
- Specification of parameters (e.g., k)

(b) Logistic Regression (8 marks)

- Implementation
- Proper handling of multiclass classification

(c) Neural Network (Optional Bonus: up to 3 marks)

(If included, capped within total 30 marks)

- Correct architecture and training
- Reasonable performance improvement or analysis

(d) Model Evaluation & Comparison (5 marks)

- Use of appropriate metrics (accuracy, confusion matrix, etc.)
- Clear comparison between models
- Insightful discussion of strengths/weaknesses

2. Dimensionality Reduction & Visualization (5 marks)

- Correct use of PCA or t-SNE
- Interpretation of clustering patterns

3. Presentation (5 marks)

- Clear explanation of methodology and results
- Logical organization and clarity